

MAT235H-5201

Vector Calculus I

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Check Quercus for more info

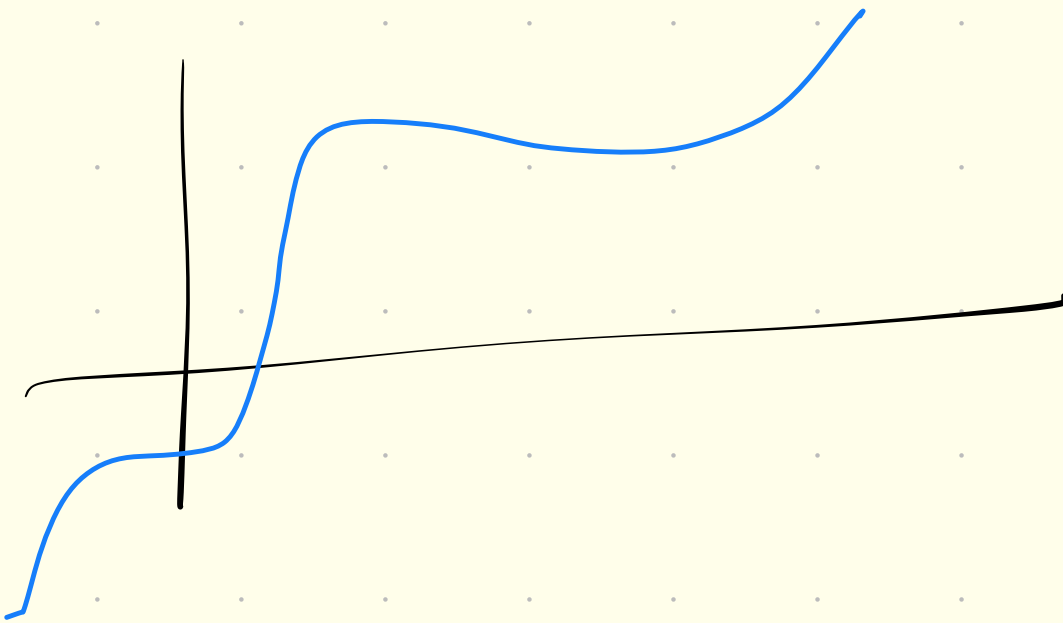
Please READ Syllabus carefully

What is multivariable calculus about?

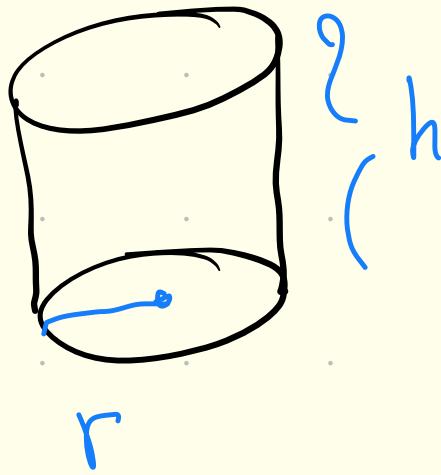
a one variable function:

Ex: polynomials; trigs, ...

x	1	2	3, 14	...
y	1	0	2	...

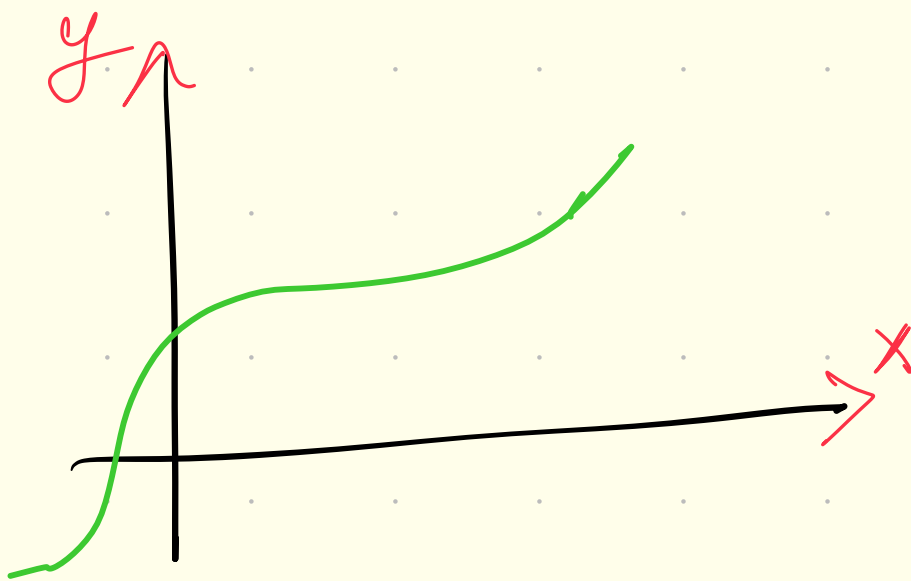


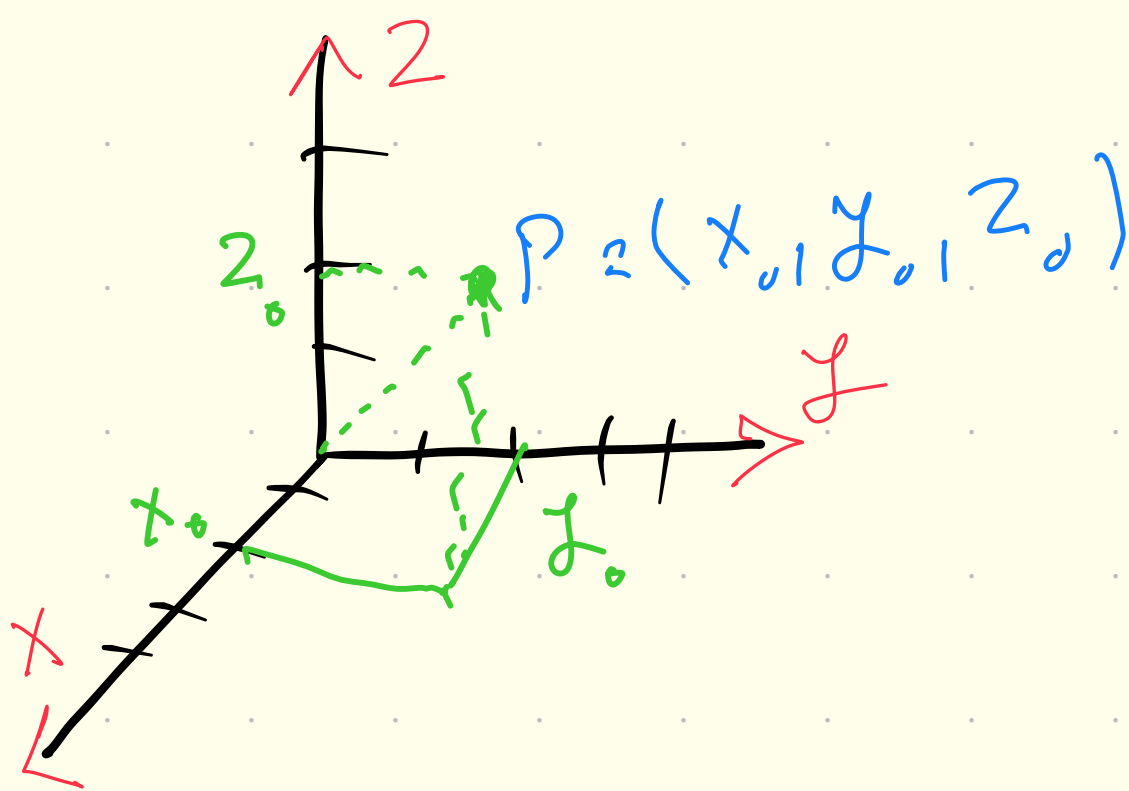
Exo



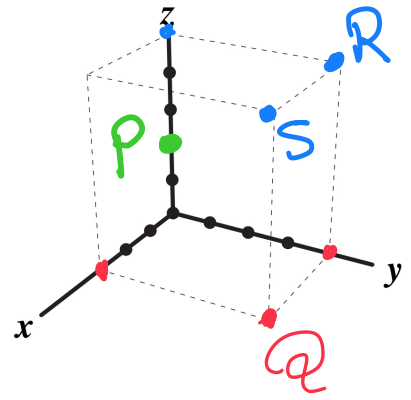
$$Vol = h \times \pi r^2$$

we need two variables

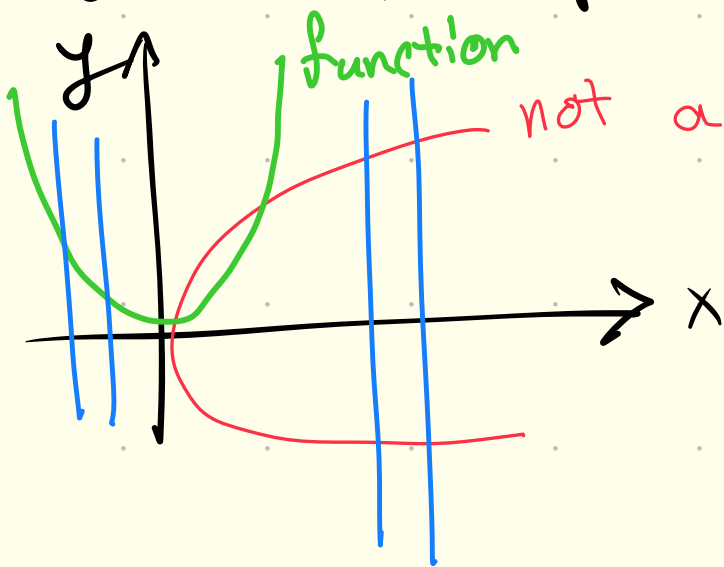




Example 1. Plot the points $P = (0, 0, 2)$, $Q = (3, 4, 0)$, $R = (0, 4, 5)$, and $S = (3, 4, 5)$ on the coordinate axes to the right.



A two variable real function is a rule to take two inputs and output one real number



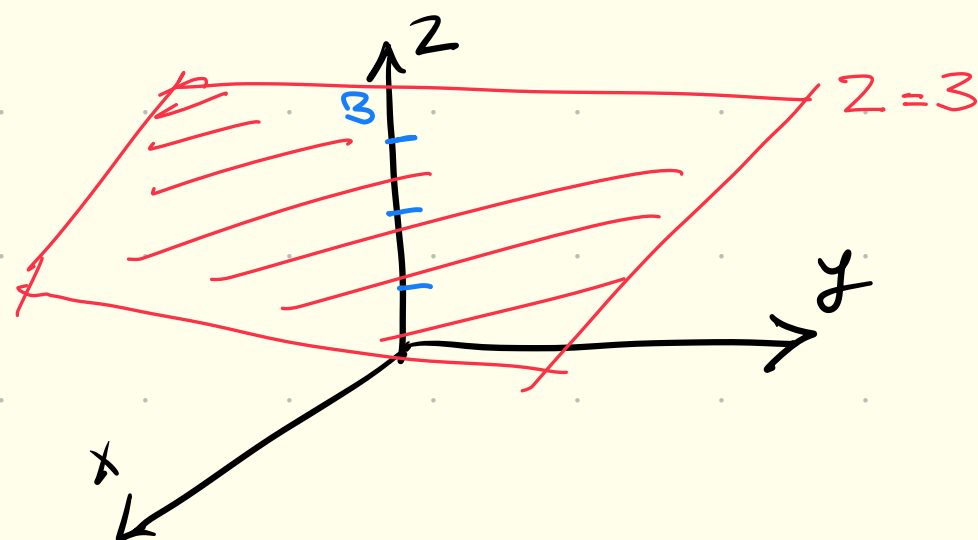
not a function (for any two inputs there is only one value)

Ex: $f(x, y) = x + y$, $g(x, y) = xy, \dots$

Example 2. Consider the functions $f(x, y) = x^2 + y^2$ and $g(x, y) = 3$. Evaluate f at several values of x and y , and sketch a graph of g .

$$f(x, y) = x^2 + y^2$$

$y \backslash x$	0	1	-1
0	0	1	1
1	1	2	2
-1	1	2	2



Example 3. Windchill temperature is a temperature which tells you how cold it feels as a result of the combination of wind and temperature. Let $C = f(w, T)$, where C is the windchill temperature (in degrees Fahrenheit) that is associated with a wind speed of w miles per hour and a temperature of T degrees Fahrenheit. A table of values for the function f is given to the right:

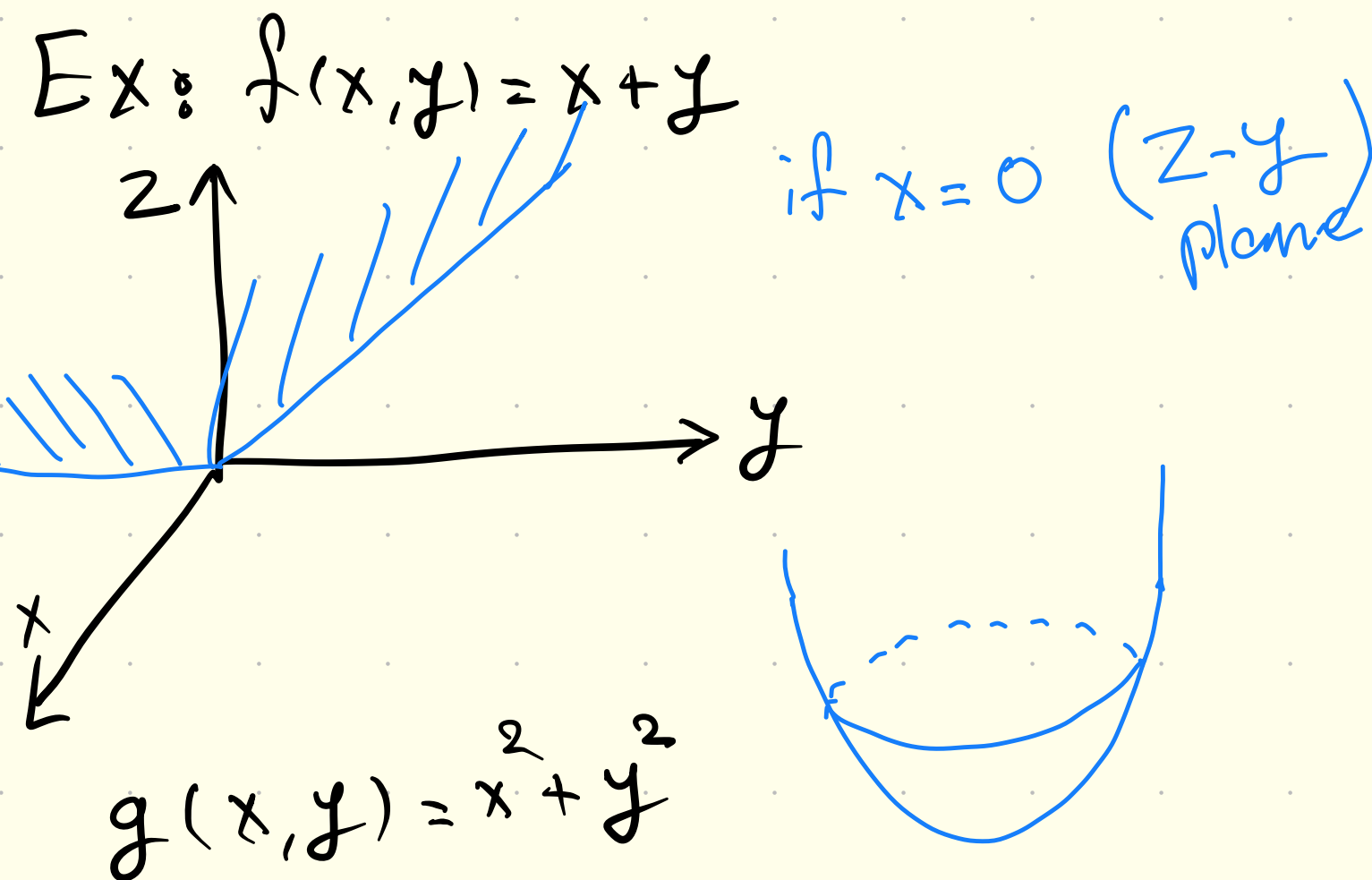
$w \setminus T$	35	30	25	20	15	10	5	0
5	31	25	19	13	7	1	-5	-11
10	27	21	15	9	3	-4	-10	-16
15	25	19	13	6	0	-7	-13	-19
20	24	17	11	4	-2	-9	-15	-22
25	23	16	9	3	-4	-11	-17	-24

- (a) Evaluate and interpret $f(20, 5)$.

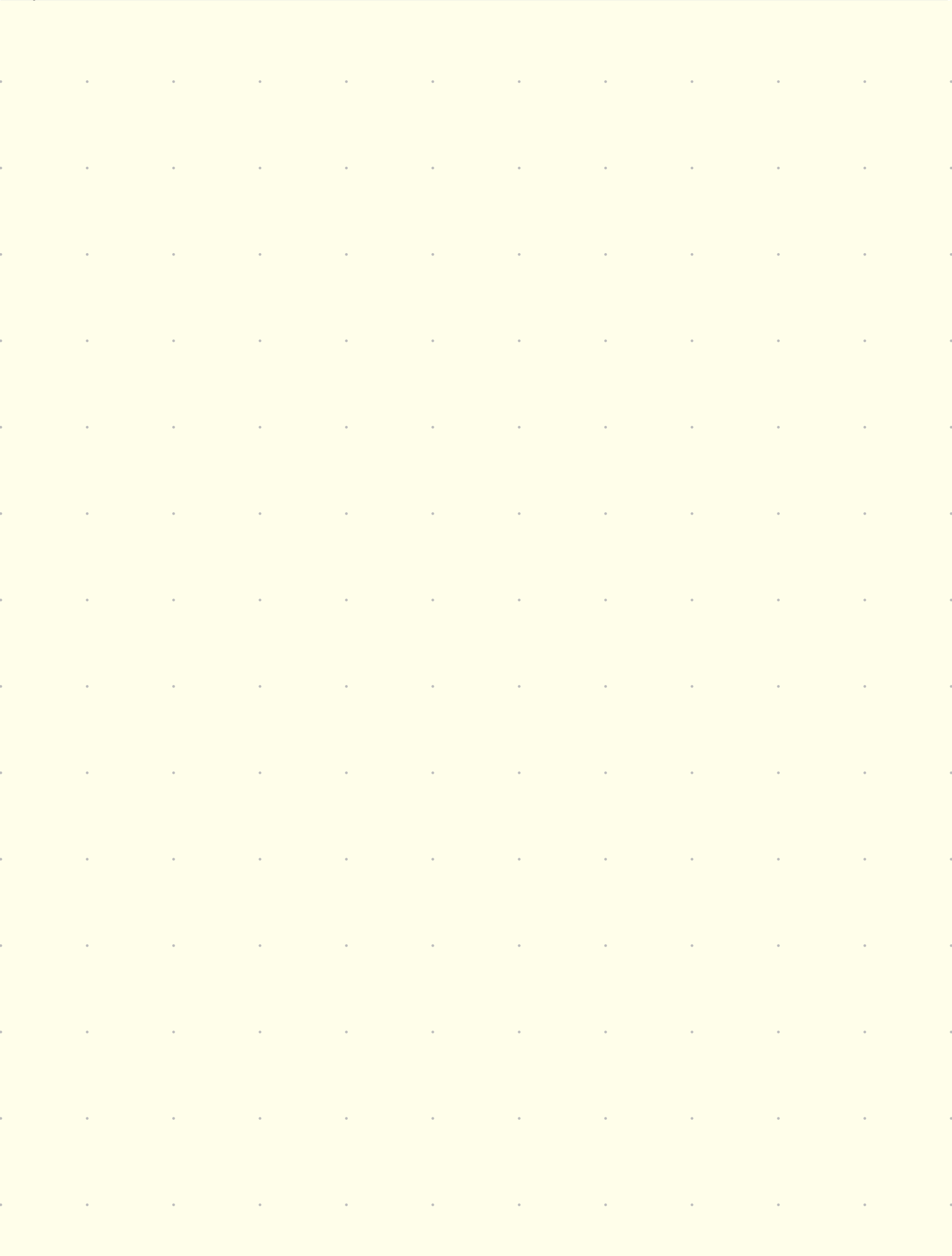
$$f(20, 5) = -15$$

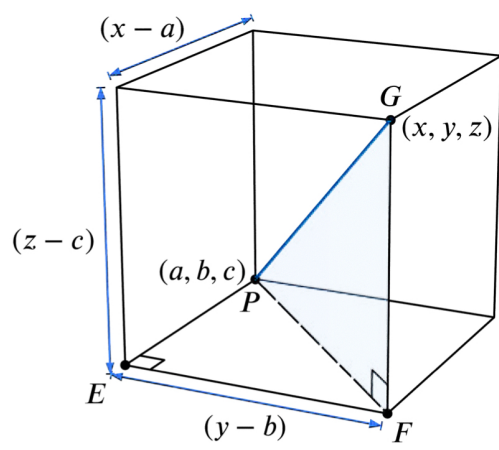
- (b) How fast does the wind need to blow for it to feel like -10°F when the air temperature is really 5°F ?

$$w \approx 10 \text{ mph}$$



Example 7 Which of the points $A = (1, -1, 0)$, $B = (0, 3, 4)$, $C = (2, 2, 1)$, and $D = (0, -4, 0)$ lies closest to the xz -plane? Which point lies on the y -axis?



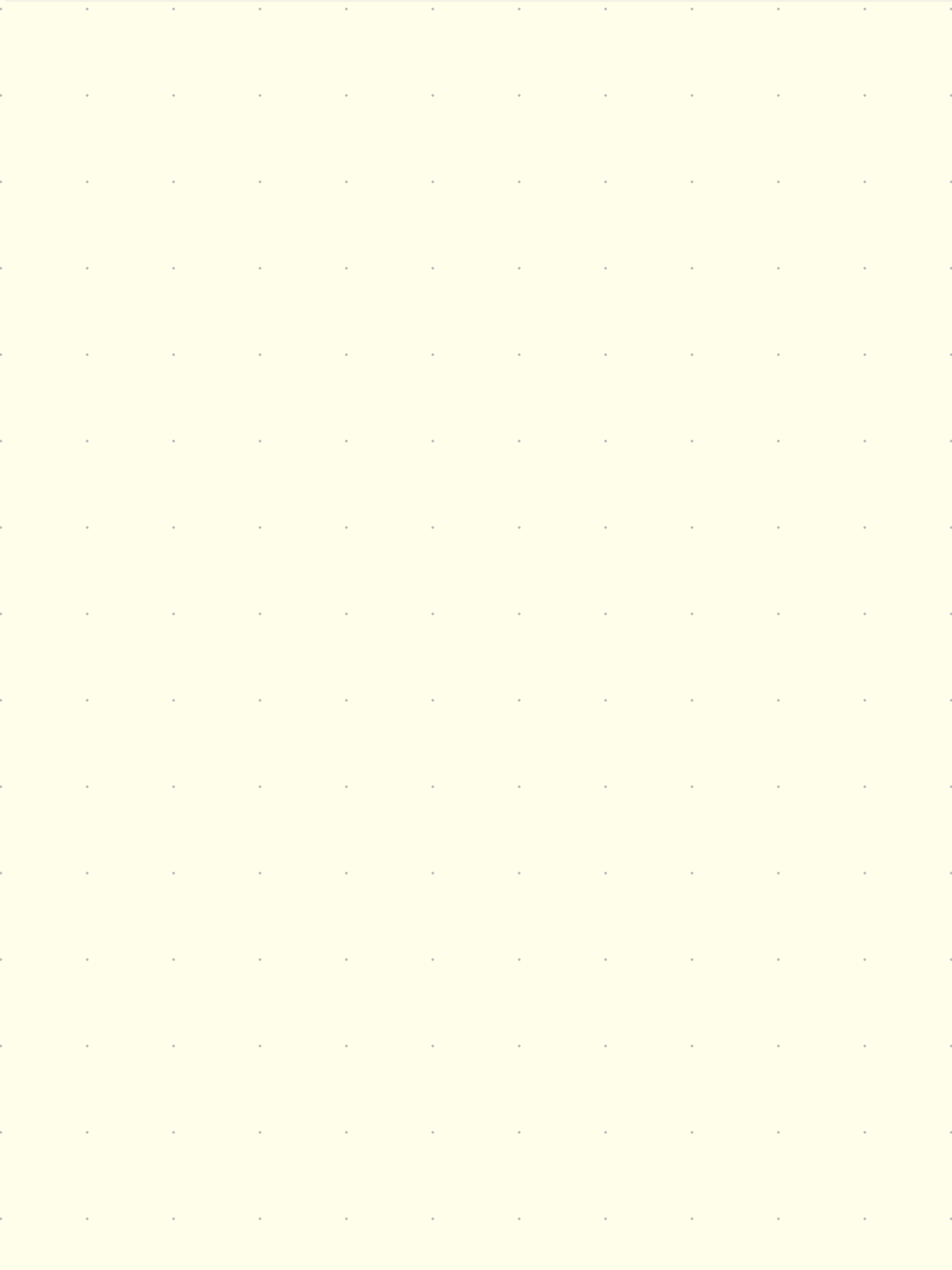


Exercises

1. Let $T = f(d, t)$, where T is the temperature, in degrees Fahrenheit, of a room t minutes after a heater is turned on, at a distance of d meters from the heater.
- (a) Is T an increasing or a decreasing function of t ? Explain. Sketch possible cross sections of f for $d = 1, 2$, and 3 meters.
- (b) Is T an increasing or a decreasing function of d ? Explain. Sketch possible cross sections of f for $t = 0, 5$, and 10 minutes.



Example 4 Describe the graph of $f(x, y) = 1 + x - y$.



Example 5

Graph the equation $x^2 + y^2 = 1$ in 3-space.

19. Consider the function f given by $f(x, y) = y^3 + xy$.
Draw graphs of cross-sections with:

- (a)** x fixed at $x = -1$, $x = 0$, and $x = 1$.
- (b)** y fixed at $y = -1$, $y = 0$, and $y = 1$.

